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HLSC 2P27 ASSIGNMENT 2

SOLO PROJECT

Critical Appraisal of Prenatal Exposure to Antibiotics and Risk of Childhood Obesity in a Multicenter Cohort Study

Overall, Purpose of the Article:

The researchers want to know whether there is a connection between Prenatal exposure to antibiotics and the risk of childhood obesity later. The researchers wanted to know if women who used antibiotics especially if they took them repeatedly or during key periods like the second trimester were more likely to have children who became overweight or obese by the age of seven. By focusing on a large cohort, the study is going to make the researchers understand the long-term metabolic effects of antibiotic use on children, this would help us understand a public health issue and make room for more research in pregnancy care and child health.

Summary of the Elements:

The study uses a prospective cohort design, using data from the Collaborative Perinatal Project (CPP) that was collected between 1959 and 1965. the researchers followed 46,000 women who gave birth to nearly 59,000 babies at 12 medical centers across the United States through their pregnancies, and their children were later tracked into childhood to monitor various health outcomes. The researchers collected information on antibiotic (penicillins, sulfonamides, tetracyclines, nitrofurans and aminoglycosides) use by asking the mothers during several prenatal visits which gave them information on the frequency and timing of antibiotic use. The timing of first exposure to antibiotics was grouped into the first trimester (month 1–3), second trimester (month 4–7), or third trimester (month 8 to delivery) of gestation. Childhood obesity was determined by measuring body mass index (BMI) percentiles and calculating BMI z-scores when the children were 4 and 7 years old. The study also used Poisson regression to estimate risk ratios and applied generalized estimating equations to get reliable standard errors.

Critical Appraisal:

When evaluating the article, the researchers successfully used a focused research question that is easy to understand. The study's purpose is to examine if prenatal exposure to antibiotics could increase the risk of childhood obesity. It is very straightforward and well-defined, which helps the readers understand the aim from the start. One of the major strengths of the study is its use of a large and comprehensive dataset. By using the CPP, the researchers had access to a reliable sample of thousands of participants, which allowed them to track long-term health outcomes. This large sample size adds a lot of credibility to the research findings by reducing the chance of random results. However, a limitation is the age of the data (1959–1965). Healthcare practices, antibiotic types, and lifestyles of people have evolved since then, thereby reducing the study's relevance to modern population.

The way the study measured antibiotic use is both a strength and a weakness. On one hand, the researchers gathered repeated reports of antibiotic use from the mothers during different stages of pregnancy, which shows exposure over time. On the other hand, the study relied on self-reported data, and even though this was done several times, it still introduces the possibility of recall bias or misclassification bias. Mothers may not accurately remember or report every instance of antibiotic use, which could lead to errors in the data and weaken the study's conclusions.

Another critical aspect to evaluate is the study's approach to measuring exposure, specifically, the use of antibiotics during pregnancy. The researchers used prescription

records as their data source, which gives an advantage over self-reported data by minimizing recall bias, However, the prescription does not confirm that the antibiotics were taken, or taken as prescribed, Even though the study does mention the general types of antibiotics used like penicillins and macrolides, it doesn't go into any real detail. We don't know how much was given, how long the mothers took them, how often, or even why they were prescribed. That kind of information really matters, because different antibiotics and reasons for using them could affect the baby in different ways. Without these details, it's difficult to understand how strong or serious the exposure really was. So even though using prescription records is a great data source, the lack of specific information makes the results less clear and hard to trust.

Furthermore, the study's focus on obesity measured at just two ages 4 and 7 which limits the ability to understand the long-term consequences of prenatal antibiotic exposure. It remains unclear whether the observations will remain present as the children grow older, perhaps into adolescence or adulthood. Additionally, the researchers did not collect detailed information on the specific infections that led to the use of antibiotics. This missing information is critical because it opens the possibility that it might not be the antibiotics themselves, but rather the infections for which they were prescribed, that contributed to the risk of developing obesity

Minor Suggestions and Comments for Improvement:

For future research, it would be better to use a more recent data that has the current healthcare practices and the modern lifestyle of pregnant women and their children and including more detailed information about the reasons behind antibiotic use. such as the type and severity of the infections, as they could help us clarify whether the drugs or the infections themselves are influencing the risk of childhood obesity. Using prescription records instead of self-reported data might also reduce the risk of recall bias and provide a more accurate measure of antibiotic exposure. Additionally, extending the follow-up period to include older children or adolescents would give a better understanding of the long-term impact of prenatal antibiotic exposure. A deep analysis that differentiates between various classes of antibiotics might also reveal whether some types have stronger effects on the risk of obesity than others. These would improve the overall quality of the research findings.

Impressions and Contributions to Research:

Overall, the article by Wang et al. makes an important contribution to the field by exploring the potential long-term effects of prenatal antibiotic exposure on childhood obesity. Despite its limitations, particularly regarding the age of the data and reliance on self-

reported data, the study provides a medium for understanding how early-life exposures can have lasting impacts on health. The huge sample size from the Collaborative Perinatal Project gives it strength, even though the data from that time doesn't totally match today's lifestyle and health care practices. The results aren't huge as there is just a small link between antibiotics and obesity. These results raise important questions about how substances moms use during pregnancy could impact their children later in life. It calls for more research using current data and utilizing prescription records instead of relying on self-reports.

REFERENCES

Wang, B., Liu, J., Zhang, Y., Yan, C., Wang, H., Jiang, F., Li, F., & Zhang, J. (2018). Prenatal Exposure to Antibiotics and Risk of Childhood Obesity in a Multicenter Cohort Study. *American Journal of Epidemiology*, *187*(10), 2159–2167. https://doi.org/10.1093/aje/kwy122